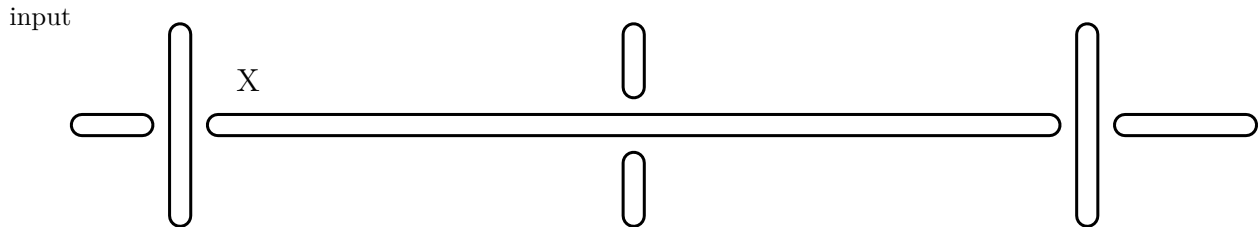
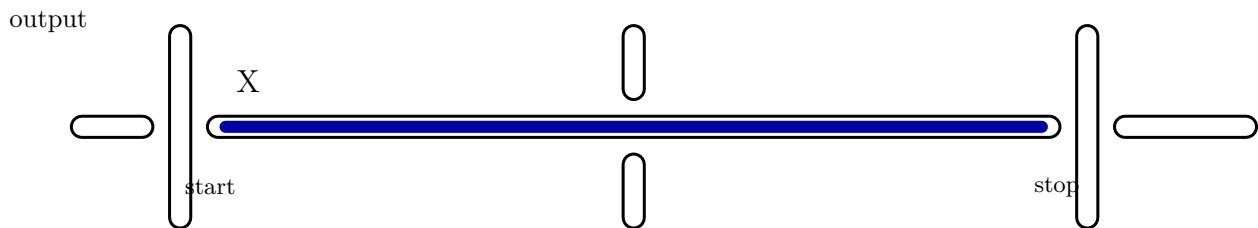


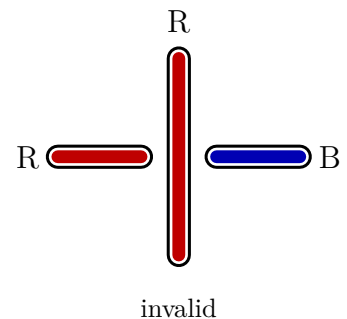
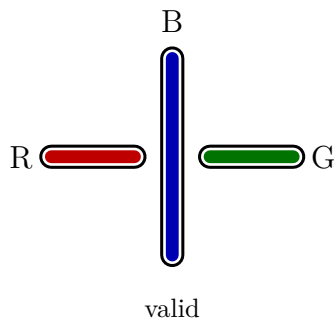
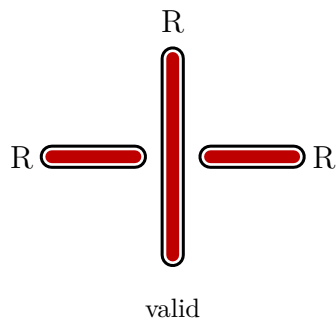
Keep cords on the table. Move them without cutting or passing strands through each other. Use red (R), blue (B), and green (G).



process: follow X across the overpass to the next gap.

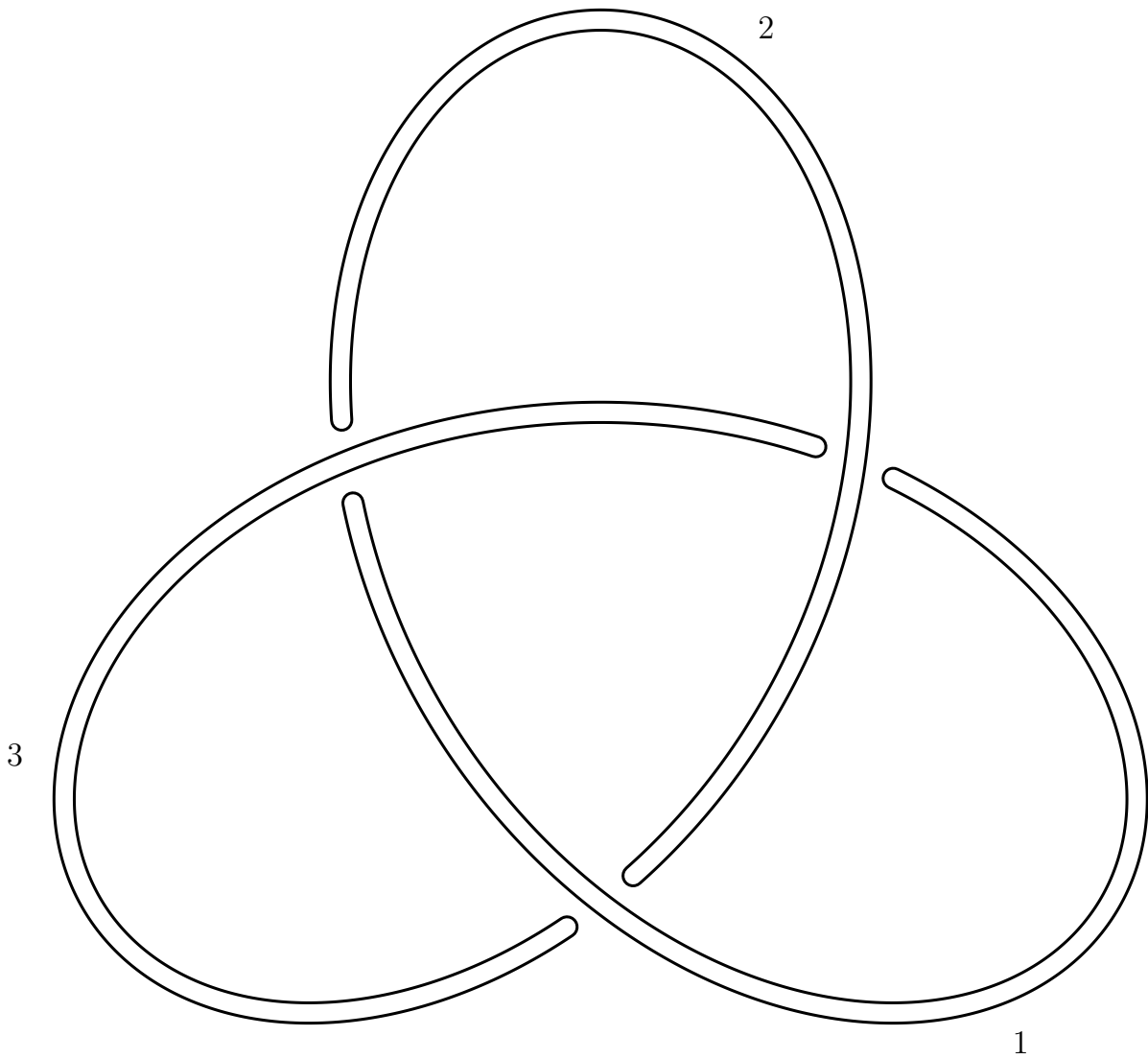


An arc runs between underpasses and continues through overpasses. A closed strand with no underpasses is one whole arc. Give each whole arc one color. At each crossing the colors must be all the same or all different.



Problem 1: Show an adult which strand goes over at each crossing above. Trace one whole arc on the next page without changing strands.

Problem 2: Find every valid coloring of this loop, including one-color choices. The numbers name whole arcs; record each coloring and explain why your list is complete.



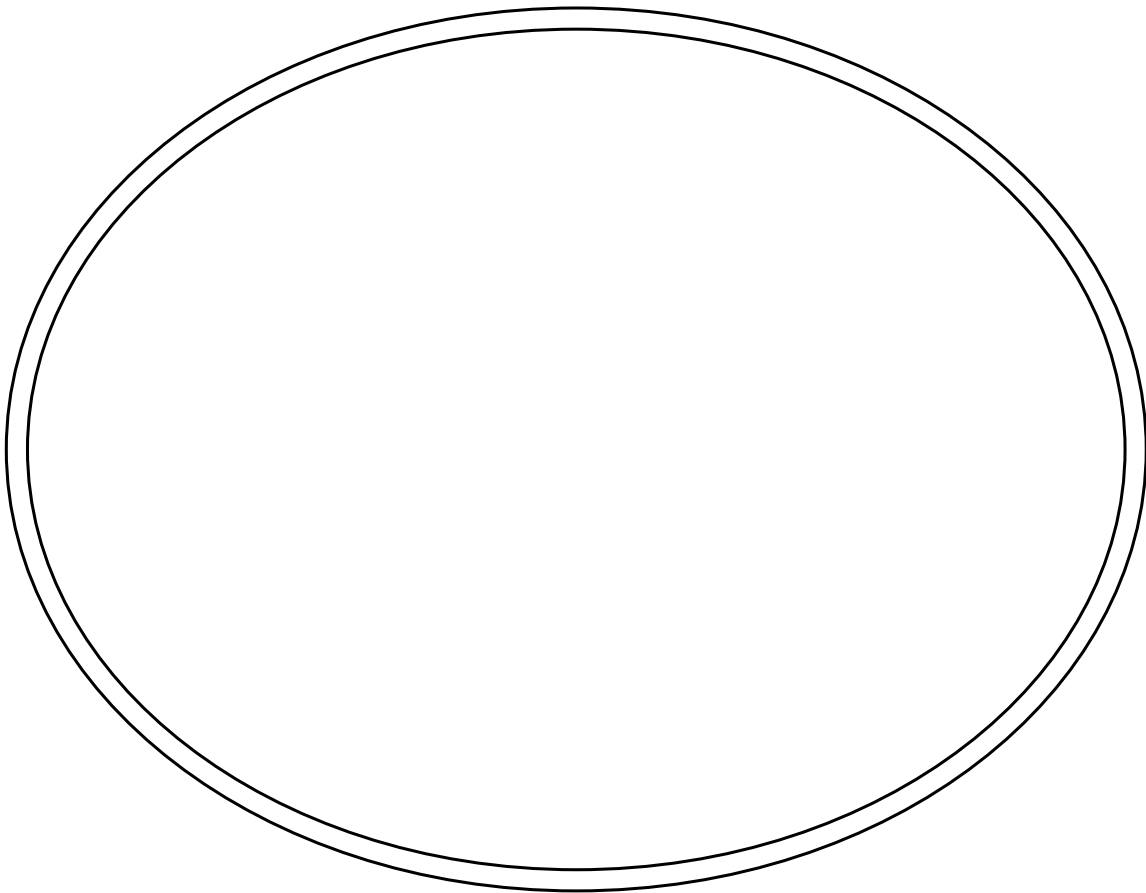
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐

1	2	3
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1	2	3
☐	☐	☐
1	2	3
☐	☐	☐

1	2	3
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1	2	3
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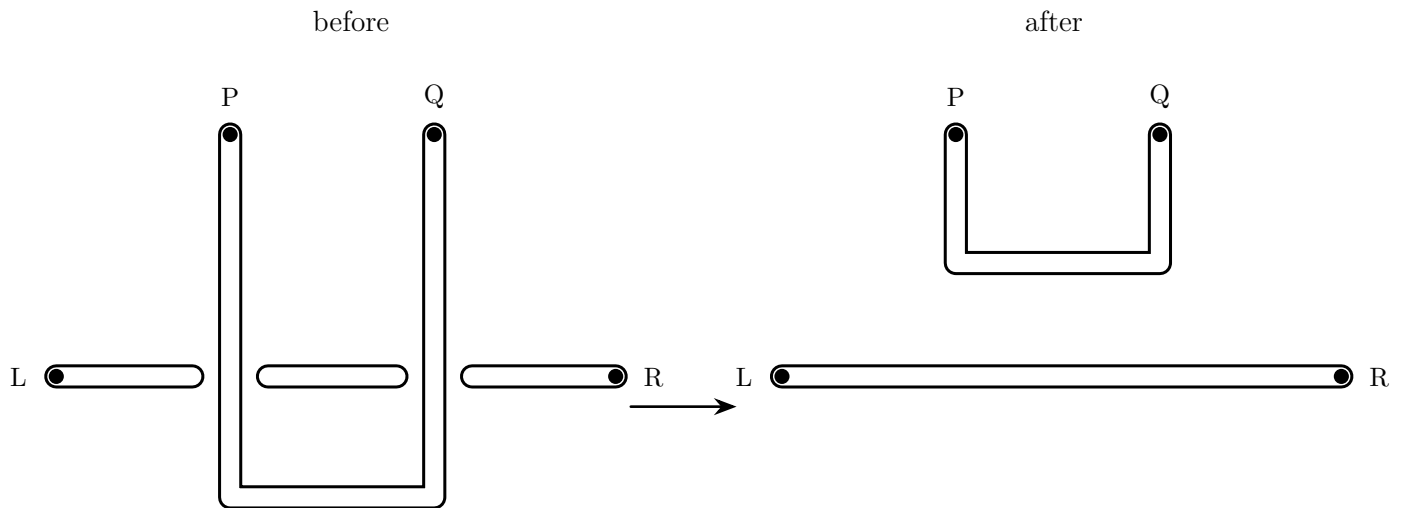
1	2	3
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1	2	3
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1	2	3
☐	☐	☐

Problem 3: Count all valid colorings of this plain loop. Compare that count with the count for the three-crossing loop.



Problem 4: Try to move each of the four adult-prepared closed cords in the tray into this plain loop shape. Separate what you managed to do from what you only suspect cannot be done.

Problem 5: An arc may stop at an end dot here. Which colors at L and P allow both pictures to be colored with matching end-dot colors? Can the same choice have more than one completion? Explain.



Problem 6: Mathematicians have shown that moving a closed loop without cutting or passing strands through each other preserves its number of valid colorings. Use your counts to decide whether the three-crossing loop can become the plain loop.

Problem 7: A knot called the figure-eight also has only three colorings, yet cannot become a plain loop. What can equal coloring counts tell you about two loops?